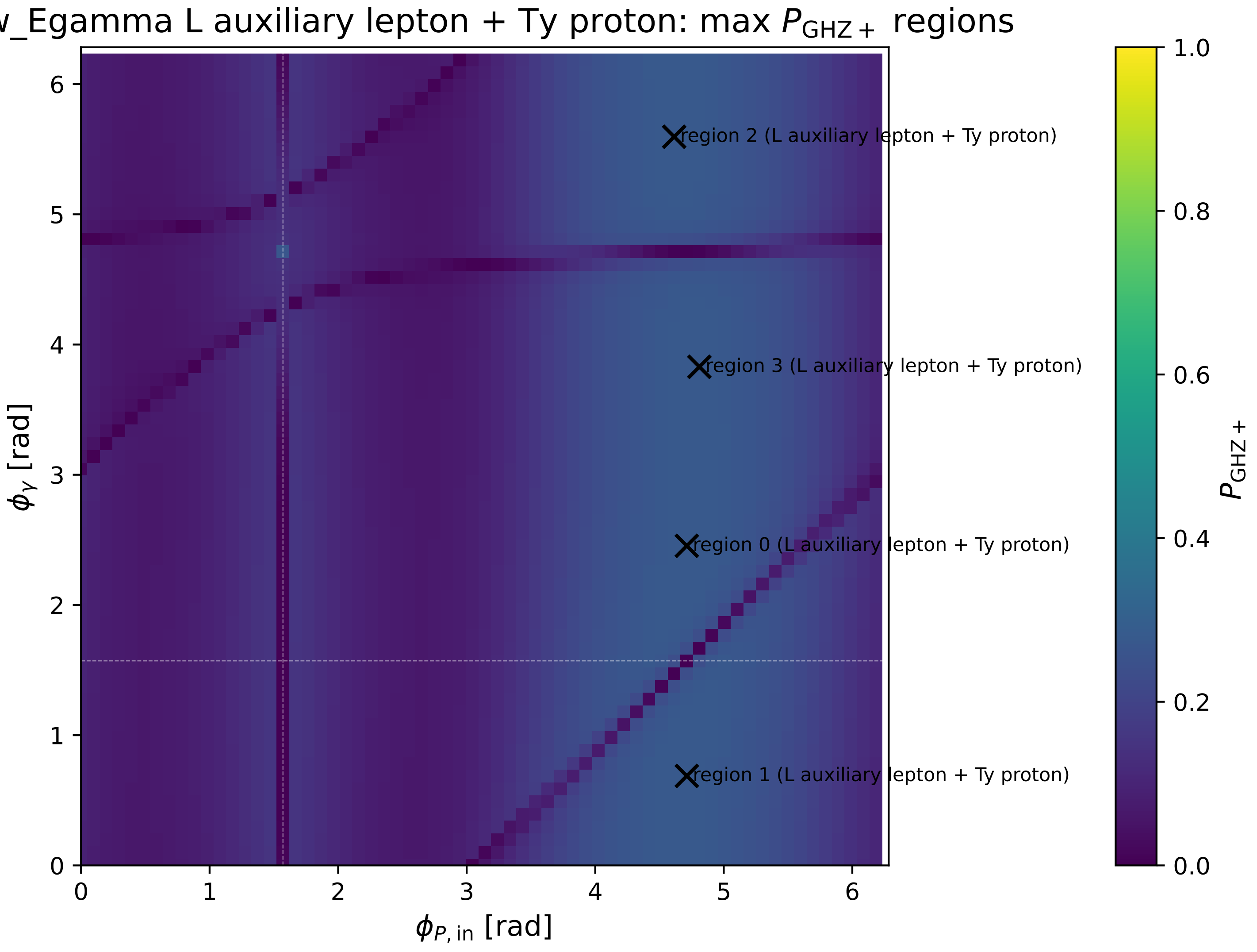
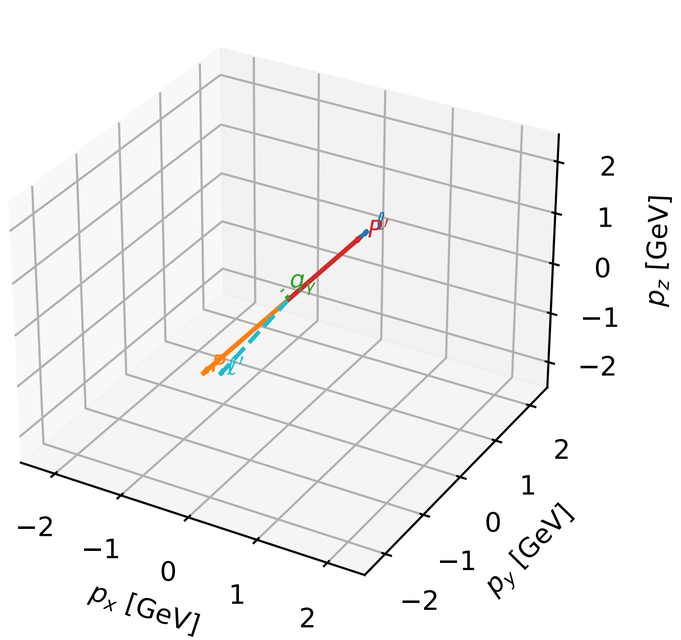

w_Egamma L auxiliary lepton + Ty proton: max $P_{\mathrm{GHZ}+}$ regions
region 2 (L auxiliary lepton + Ty proton)
region 3 (L auxiliary lepton + Ty proton)
region 0 (L auxiliary lepton + Ty proton)
region 1 (L auxiliary lepton + Ty proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$P_{\mathrm{GHZ}+}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

L auxiliary lepton + Ty proton characteristic max $P_{\mathrm{GHZ}+}$ configuration

3D momenta

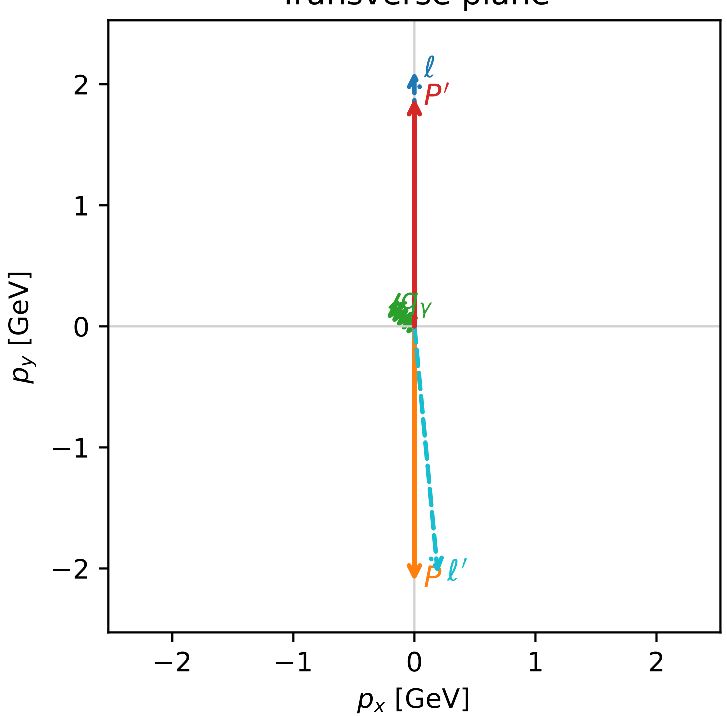


ghz_purity_low_egamma_lty_region_0 P_{\rm GHZ+}=0.276902
region: low_Egamma_LTy_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_\ell\rangle \otimes |T_{y,p}\rangle$

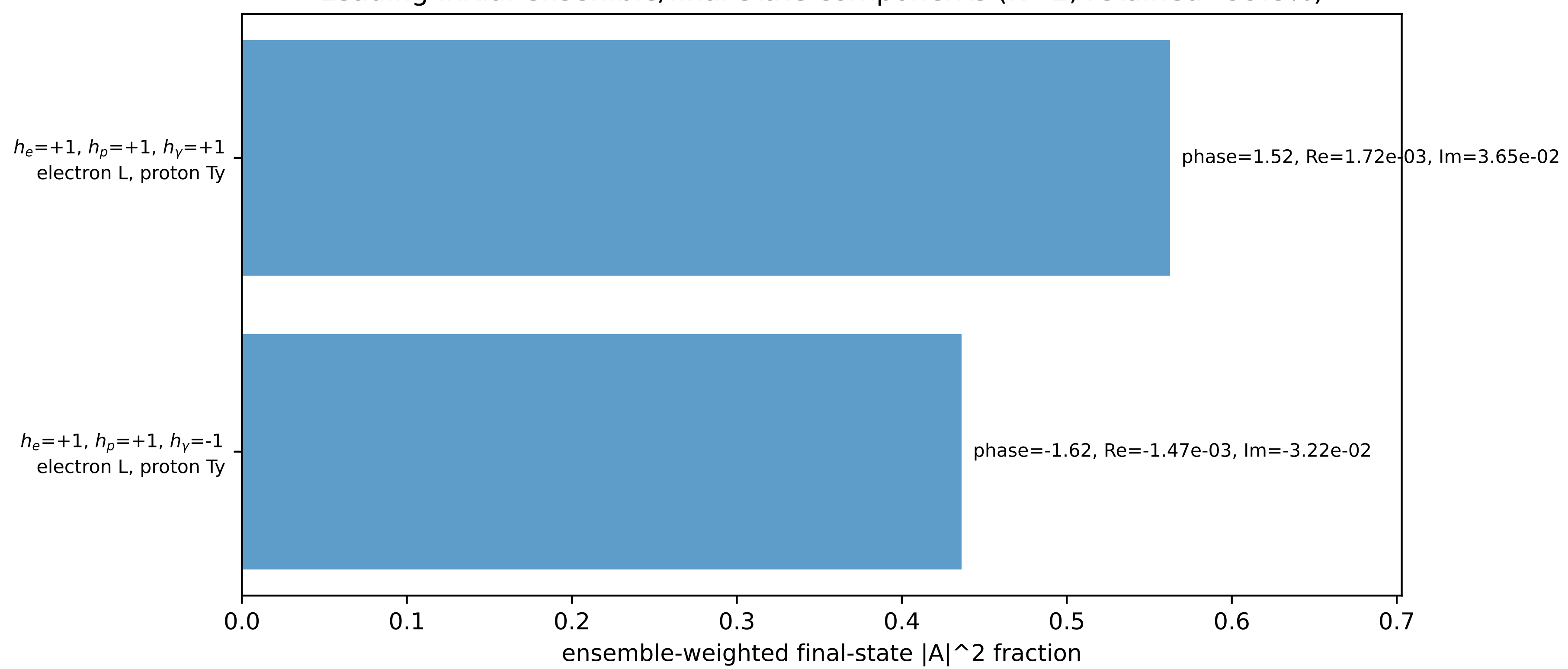
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=1.88483
θ_{in}=1.57079, ϕ_ℓ=1.5708, ϕ_P=4.71239
E_γ=0.25, ϕ_γ=2.45437
Q^2=17.2605, x_B=0.974327, t=-15.8938, W^2=1.33465, y=0.902185
$\theta(\ell', \gamma)$=134.778 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= 0.0000 p_y= 2.1069 p_z= -0.0000
P E= 2.3063 p_x= -0.0000 p_y= -2.1069 p_z= 0.0000
ℓ' E= 2.2832 p_x= 0.1933 p_y= -2.0434 p_z= 0.0000
P' E= 2.1053 p_x= 0.0000 p_y= 1.8848 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.1933 p_y= 0.1586 p_z= 0.0000

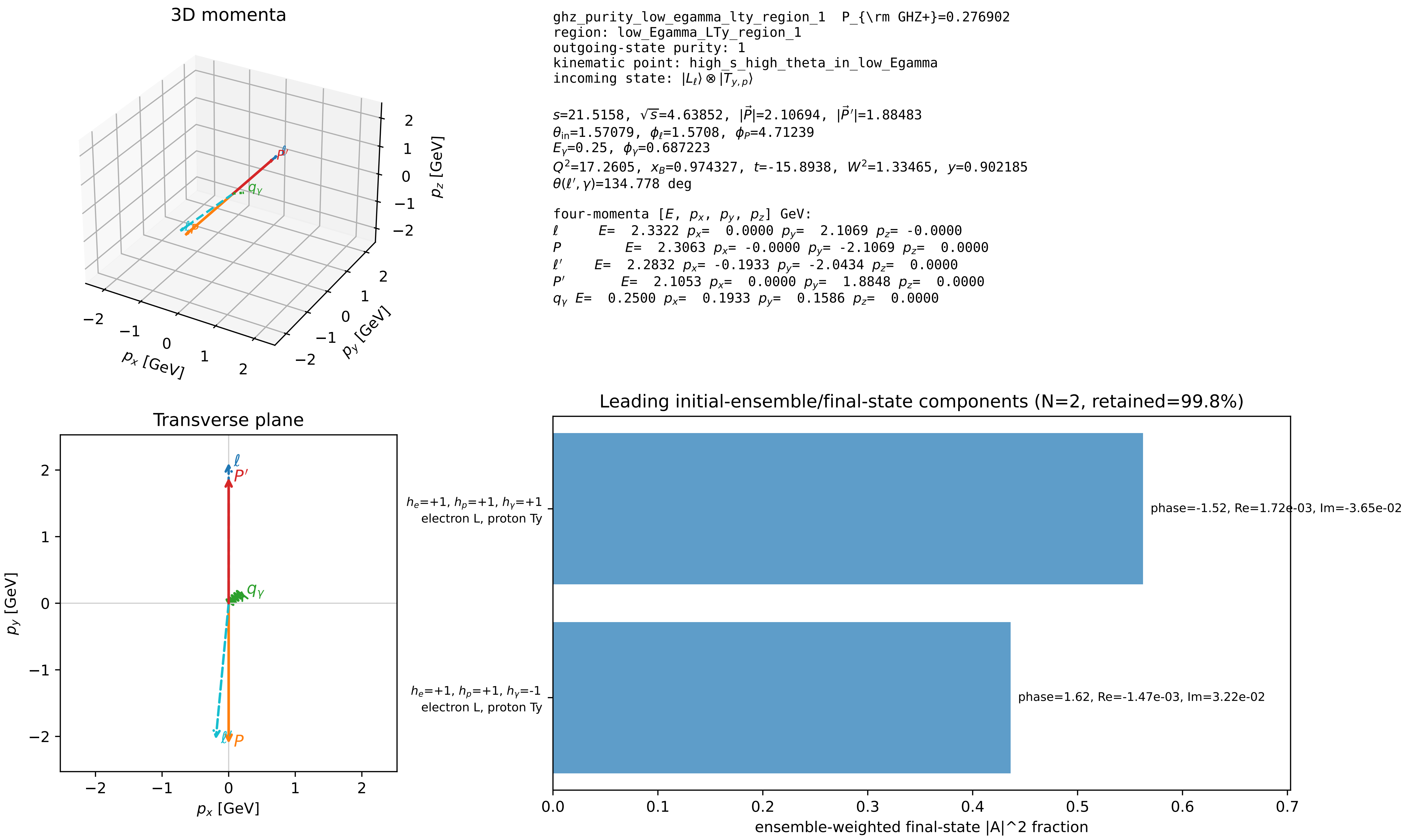
Transverse plane



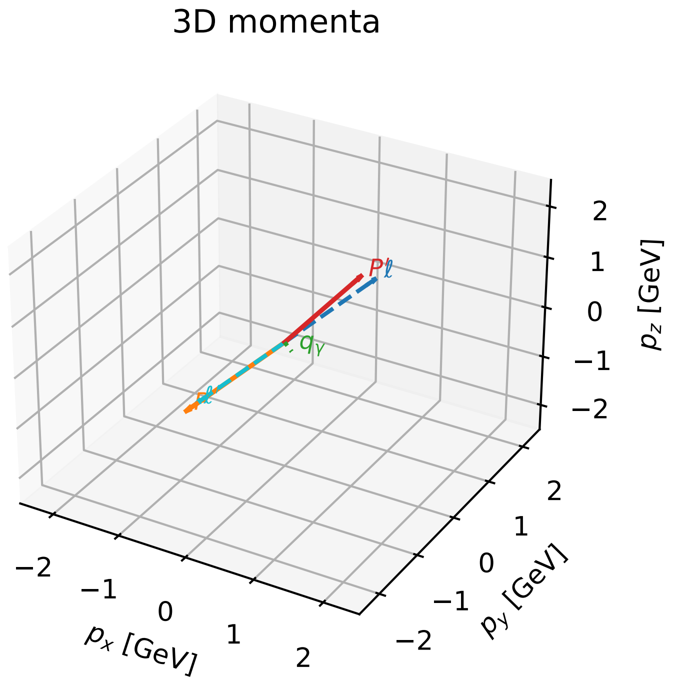
Leading initial-ensemble/final-state components (N=2, retained=99.8%)



L auxiliary lepton + Ty proton characteristic max $P_{\rm GHZ+}$ configuration



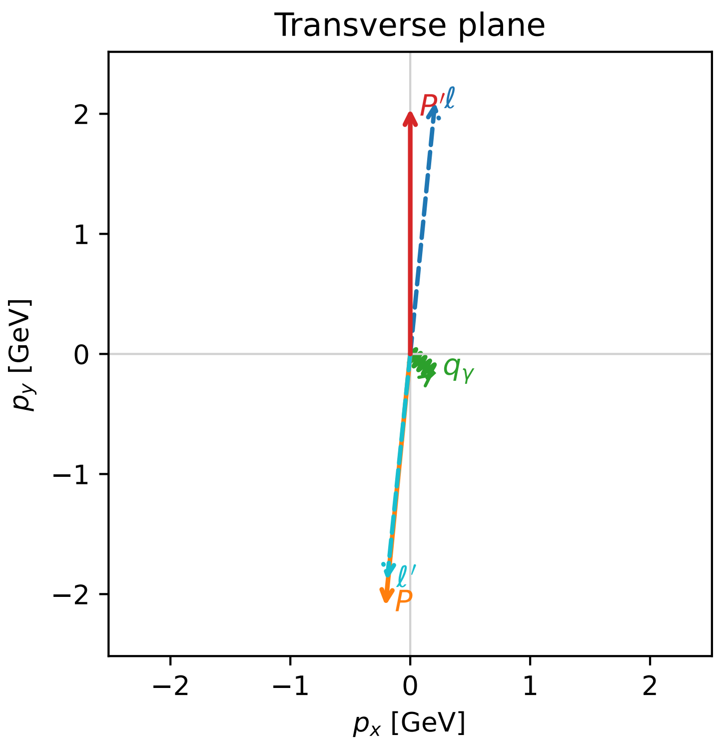
L auxiliary lepton + Ty proton characteristic max $P_{\rm GHZ+}$ configuration

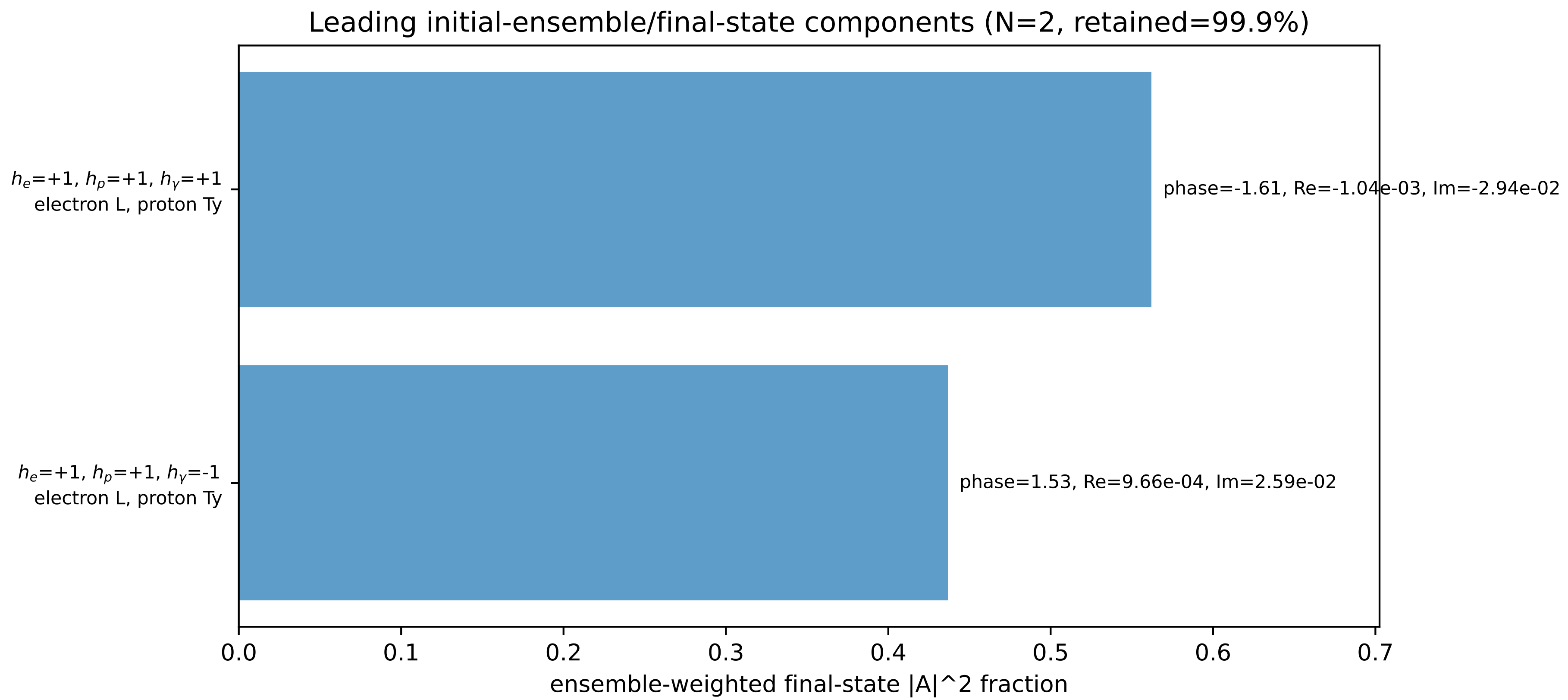


ghz_purity_low_egamma_lty_region_2 P_{\rm GHZ+}=0.276277
region: low_Egamma_LTy_region_2
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_\ell\rangle \otimes |T_{y,p}\rangle$

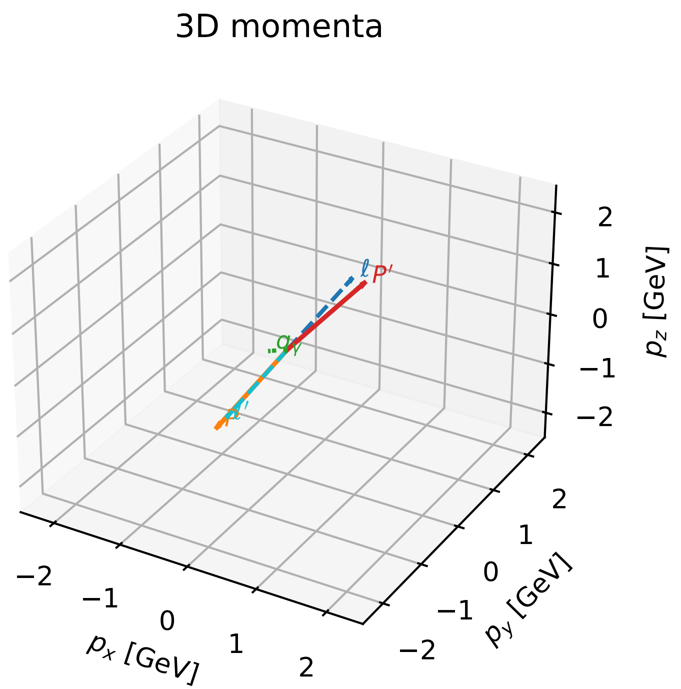
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=2.04213
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.47262, ϕ_P=4.61421
E_γ=0.25, ϕ_γ=5.59596
Q^2=15.9664, x_B=0.900135, t=-17.1699, W^2=2.65122, y=0.903327
$\theta(\ell', \gamma)$=56.4831 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= 0.2065 p_y= 2.0968 p_z= -0.0000
P E= 2.3063 p_x= -0.2065 p_y= -2.0968 p_z= 0.0000
ℓ' E= 2.1413 p_x= -0.1933 p_y= -1.8835 p_z= 0.0000
P' E= 2.2472 p_x= 0.0000 p_y= 2.0421 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.1933 p_y= -0.1586 p_z= 0.0000





L auxiliary lepton + Ty proton characteristic max $P_{\rm GHZ+}$ configuration



ghz_purity_low_egamma_lty_region_3 P_{\rm GHZ+}=0.276277
region: low_Egamma_LTy_region_3
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_\ell\rangle \otimes |T_{y,p}\rangle$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.10694, $|\vec{P}'|$=2.04213
$\theta_{\rm in}$=1.57079, ϕ_ℓ=1.66897, ϕ_P=4.81056
E_γ=0.25, ϕ_γ=3.82882
Q^2=15.9664, x_B=0.900135, t=-17.1699, W^2=2.65122, y=0.903327
$\theta(\ell', \gamma)$=56.4831 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.3322 p_x= -0.2065 p_y= 2.0968 p_z= -0.0000
P E= 2.3063 p_x= 0.2065 p_y= -2.0968 p_z= 0.0000
ℓ' E= 2.1413 p_x= 0.1933 p_y= -1.8835 p_z= 0.0000
P' E= 2.2472 p_x= 0.0000 p_y= 2.0421 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.1933 p_y= -0.1586 p_z= 0.0000